

Benjamin Garcia

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EDUCATION

University of California, Los Angeles

Bachelor of Science, Computer Science

Los Angeles, CA

Expected Graduation: Jun. 2027

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, C, C++, SQL, JavaScript, HTML/CSS

Frameworks & Tools: Next.js, React, Tailwind CSS, FastAPI, Prisma, Node.js, Redis, PostgreSQL, Supabase, Firestore, JIRA

Cloud & DevOps: AWS Lambda, Palantir Foundry, Docker, Vercel, Git, CI/CD, Linux/Unix, GitLab, Kubernetes

EXPERIENCE

Undergraduate Research Developer

Dec. 2025 – Present

UCLA Human-Computer Interaction Research Lab

Los Angeles, CA

- Building interactive prototypes in **React**, **TypeScript**, and **Next.js** supporting research on human-AI collaboration, creativity systems, and interactive data exploration.
- Collaborating with **PhD researchers** to design, implement, and refine UI systems used in user studies, experiment tooling, and demos for upcoming **CHI/IUI/NAACL** submissions.
- Integrating **LLM-powered features** via Python APIs and real-time channels such as **WebSockets**, enabling rapid iteration cycles aligned with research hypotheses.

Software Engineer Intern

Mar. 2025 – Sep. 2025

Todd Agriscience

Remote, Part-time

- Delivered AI-driven crop insight dashboards for **5–10** clients by architecting full-stack systems in **Next.js**, **Tailwind CSS**, and **Framer Motion**, integrated with **Palantir Foundry** and **AWS Lambda**.
- Developed dynamic data pipelines visualizing **20+** agricultural data streams (soil, irrigation, weather, yield) to enable real-time farm management decisions.

Research and Development Engineer Intern

Jul. 2025 – Aug. 2025

Bonterra

Remote

- Prototyped an **agentic AI** system for nonprofit event analytics using **tool-calling frameworks** and **Model Context Protocols (MCPs)**, modeling attendance, retention, and fundraising across datasets; benchmarked **5+** agent frameworks.
- Developed and tested a multi-tool orchestration layer enabling **autonomous agent collaboration**, improved retrieval latency by **22%**, and validated tool-selection policies for scaling across nonprofit datasets.
- Presented findings to **40–50** cross-functional staff across engineering, analytics, and leadership, influencing early **machine-learning** adoption in nonprofit strategy.

Software Engineer Intern

May 2025 – Jun. 2025

TensorStar

Remote

- Engineered real-time agentic UIs with **Next.js**, **Redux**, and **WebSockets** in **Agile sprints**, supporting **100+** concurrent users with sub-**100ms** latency.
- Built secure credential-submission workflows integrated with **HashiCorp Vault** and a **Python + Redis** backend, unifying authentication across 50+ enterprise data sources (AWS Glue, Snowflake, MongoDB, Postgres, Airflow, dbt, Spark).
- Contributed **30+** reusable UI components and coordinated with a **10–15-engineer** cross-functional team spanning frontend, ML, and infrastructure.

PROJECTS

Poly Clubs | *Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS*

Dec. 2025 – Present

- Built a full-stack review platform for **493 student organizations** using **Next.js 15** and **Supabase**, implementing browser fingerprinting and database-level rate limiting to prevent spam while maintaining anonymous reviews.
- Designed a serverless moderation system with **PostgreSQL RPC functions** and **Row Level Security policies**, reducing client-server round trips by **60%** through database-level aggregation and achieving **10-100x** faster query performance via strategic indexing.
- Architected security-first input validation with **multi-layer constraints** (client, server, database), integrated **FingerprintJS** for cross-session rate limiting, and implemented **dynamic OpenGraph metadata generation** for improved SEO across 493 statically-generated pages.

Git Proof | *Next.js, FastAPI, Python, Redis, Celery, Gemini 2.0, PostgreSQL*

May 2025 – Nov. 2025

- Built a **full-stack analytics platform** that transforms raw GitHub activity into recruiter-ready insights, using **FastAPI** and **PostgreSQL** as the core backend and data layer.
- Developed an **AI-driven analysis layer** with **Gemini 2.0 Flash-Lite** to generate automated summaries of coding habits and repository READMEs, leveraging **Redis** caching to minimize latency for repeated dashboard access.
- Engineered a **scalable background processing system** with **Celery** and **Redis-backed task queues**, incorporating idempotent jobs and retry logic to reliably ingest and process large repositories.